

ApexPlanet Data Analytics Internship

Final Project Report

1. Project Overview

This report summarizes the work completed during the ApexPlanet Data Analytics Internship. The project focused on analyzing the UCI Online Retail dataset, which contains transnational transactions for a UK-based online retail store. The core objectives included data cleaning, SQL extraction, statistical analysis, machine learning (clustering and predictive modeling), and the deployment of an automated data pipeline.

2. Data Cleaning & Feature Engineering (Task 1 & 5)

The raw dataset contained 541,909 records. A rigorous data cleaning process was applied:

- Duplicate rows were removed.
- Cancelled orders (InvoiceNo starting with 'C') were filtered out.
- Erroneous records with zero or negative Quantity and UnitPrice were removed.
- Missing CustomerID values were handled by dropping incomplete records.
- Feature engineering was applied to create 'TotalSales' (Quantity * UnitPrice) and time-series features (Year, Month, Day, Week).

After cleaning, the final processed dataset contained 392,693 pristine records.

3. SQL Data Extraction (Task 2)

Using SQLite, the dataset was queried to extract foundational and advanced business insights. Queries were designed to identify top-performing product categories, calculate monthly recurring revenue, and determine the distribution of customer lifetime value (CLV).

4. Statistical Analysis & Clustering (Task 4)

Advanced analytics were applied to uncover hidden patterns within customer behaviors:

- K-Means Clustering: Customers were segmented based on their Recency, Frequency, and Monetary (RFM) scores to identify VIP customers, loyalists, and at-risk demographics.
- Statistical Inference: Hypothesis testing was used to evaluate differences in purchasing habits across various geographical regions.
- Predictive Modeling: A Linear Regression model was built to predict future customer spend based on historical activity, yielding measurable performance via R-squared and RMSE metrics.

5. Automation Pipeline & Business Insights (Task 5)

A robust, end-to-end Python automation pipeline was deployed to process raw data consistently. The pipeline loads data, executes the cleaning rules, calculates Key Performance Indicators (KPIs), and exports an automated multi-sheet Excel report.

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Key Business KPIs Derived:

- Total Revenue: £ 8,887,209
- Total Orders: 18,532
- Total Units Sold: 5,152,002
- Unique Customers: 4,338
- Unique Products: 3,665
- Average Order Value (AOV): £ 479.56
- Average Units per Order: 278